YatraGuard

Intelligent Safety-Centric Urban Navigation & Emergency Response System

ABSTRACT

Urban mobility in rapidly growing cities presents safety challenges, particularly for women, college students, late-night shift workers, and vulnerable commuters. Conventional navigation applications mainly prioritize distance and travel time, often overlooking poorly lit streets, isolated pathways, crime-prone areas, and other safety risks. YatraGuard is an intelligent, context-aware navigation and personal safety platform designed to consider commuter safety alongside travel efficiency. The system uses a multi-factor Safety Scoring Engine to evaluate road segments on a scale of 1 to 100 using available indicators such as street lighting, pedestrian density, CCTV coverage, police patrol proximity, historical crime information, time of day, and community-reported hazards. It offers three route choices—Safest, Balanced, and Fastest—so users can compare safety considerations with travel time. A Safety Heatmap highlights potentially risky areas, while landmark-based navigation supports familiar searching in Indian urban environments. YatraGuard includes an Emergency SOS System for rapid alerts and live-location sharing through supported channels, along with an audio siren intended to attract nearby attention. Community hazard reporting enables users to report faulty streetlights, isolated pathways, and harassment-related concerns. A credibility-based approach helps distinguish verified information from unconfirmed reports. A lightweight mode is also designed to support budget smartphones and low-bandwidth networks. The proposed technology stack includes TypeScript and JavaScript for application development, React 19 and TanStack Start/Router for the frontend, Tailwind CSS and Radix UI for the interface, and Leaflet with React-Leaflet for mapping and geospatial display. Browser capabilities such as the HTML5 Geolocation API, Web Audio API, and Vibration API support location awareness and emergency interactions. Vite supports bundling, while Cloudflare Pages/Workers compatibility supports deployment. By combining geospatial analysis, safety scoring, route optimization, community reporting, and emergency assistance, YatraGuard aims to encourage proactive risk awareness and safer route selection. Its effectiveness depends on the availability, accuracy, and timely updating of safety data; risk scores provide guidance and cannot guarantee personal safety.

KEYWORDS

Safe Navigation, TypeScript, JavaScript, Safety Scoring, Route Optimization, Geospatial Analysis, Emergency SOS, Safety Heatmap, Community Reporting.